

University of Groningen

Validity of maximum performance tasks to diagnose motor speech disorders in children

Thoonen, Geert; Maassen, B.; Gabreëls, F.; Schreuder, R.

Published in:
Clinical Linguistics and Phonetics

DOI:
[10.1080/026992099299211](https://doi.org/10.1080/026992099299211)

IMPORTANT NOTE: You are advised to consult the publisher's version (publisher's PDF) if you wish to cite from it. Please check the document version below.

Document Version
Publisher's PDF, also known as Version of record

Publication date:
1999

[Link to publication in University of Groningen/UMCG research database](#)

Citation for published version (APA):

Thoonen, G., Maassen, B., Gabreëls, F., & Schreuder, R. (1999). Validity of maximum performance tasks to diagnose motor speech disorders in children. *Clinical Linguistics and Phonetics*, 13(1), 1-23.
<https://doi.org/10.1080/026992099299211>

Copyright

Other than for strictly personal use, it is not permitted to download or to forward/distribute the text or part of it without the consent of the author(s) and/or copyright holder(s), unless the work is under an open content license (like Creative Commons).

The publication may also be distributed here under the terms of Article 25fa of the Dutch Copyright Act, indicated by the "Taverne" license. More information can be found on the University of Groningen website: <https://www.rug.nl/library/open-access/self-archiving-pure/taverne-amendment>.

Take-down policy

If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

Downloaded from the University of Groningen/UMCG research database (Pure): <http://www.rug.nl/research/portal>. For technical reasons the number of authors shown on this cover page is limited to 10 maximum.

Validity of maximum performance tasks to diagnose motor speech disorders in children

G. THOONEN, B. MAASSEN

Child Neurology Centre/Institute of Medical Psychology, University Hospital Nijmegen, Nijmegen, The Netherlands

F. GABREËLS

Child Neurology Centre, University Hospital Nijmegen, The Netherlands

R. SCHREUDER

Interfaculty Research Unit for Language and Speech, University of Nijmegen, The Netherlands

(Received 18 July 1997; accepted 14 January 1998)

Abstract

In a previous study a diagnostic procedure was developed to assess motoric involvement in clear cases of developmental apraxia of speech (DAS) and spastic dysarthria, as well as normal-speaking children. The aim of the present study is to cross-validate this diagnostic procedure. To achieve this aim the maximum performance tasks of the protocol, consisting of maximum vowel and fricative prolongation, and maximum syllable repetition, were administered to less clear cases of DAS and spastic dysarthria, children with a speech disorder of unknown origin and newly selected normal-speaking children (total number of children in the study, 72; aged predominantly between 4 and 12 years). The results showed that spastic dysarthria can be diagnosed on the basis of the maximum rate of repetitive sequences ('papa . . .', 'tata . . .', 'kaka . . .') in combination with maximum vowel prolongation. DAS can be diagnosed on the basis of the maximum rate of alternating sequences ('pataka . . .') in combination with maximum fricative prolongation. Using the diagnosis of speech-language pathologists as a criterion, sensitivity and specificity values were obtained ranging from 89 to 100%. Among the children with a speech disorder of unknown origin significant dysarthric or apraxic involvement was observed. Thus, it can be concluded that the diagnostic procedure yields quantitative measures of the degree to which dysarthria or apraxia plays a role in the development and maintenance of speech disorders in children.

Keywords: motor speech disorders in children, assessment, maximum performance tasks, validity.

Introduction

Estimates of the prevalence of speech disorders in children range between 2.5 and 10% of the normal school population, i.e. the population of children who do not

Address correspondence to: Geert Thoonen, PhD, Centre of Child Neurology (I.K.N.C.), University Hospital Nijmegen, P.O. Box 9101, 6500 HB Nijmegen, The Netherlands.

require medical or special educational treatment because of cognitive deficits, hearing impairment, craniofacial dysmorphology or neuromotor dysfunction (Enderby and Philipp, 1986; Shriberg, Kwiathowski, Best, Hengst and Terselic-Weber, 1986; Shriberg, 1994; Dodd, 1995). Even after exclusion of the latter impairments a heterogeneous group of children remains, which encompasses (functional) articulation disorders, phonological delay, (mild) developmental dysarthria, and developmental apraxia of speech. To date there exists no generally accepted classification system, although several systems have been proposed (Shriberg and Kwaithowski, 1982; Dodd, 1995). In a recent review of the literature on the problems of differential diagnosis in a clinical setting, Morgan Barry (1995) showed that at the level of speech symptoms there is much overlap between the different conditions. Moreover, in many cases aetiological data are not available, or not sufficiently specific, to serve as a basis for diagnostic differentiation.

For clinicians to facilitate appropriate treatment the speech diagnosis needs to contain a theoretically based judgement on the deficits underlying the speech impairment (Schwartz, 1992; Bradford and Dodd, 1996). This clinical perspective calls for a more eclectic system of classifying speech disorders, making use of whatever information is available with respect to symptomatology and possibly available information with respect to aetiology, treatment history and educational and psychological variables (Dodd, 1995; Bradford and Dodd, 1996). Therefore, as an alternative for the above-mentioned typological classification systems, a diagnostic system that is based on the recognition of the underlying deficit, or level of breakdown (Dodd, 1995), would be most useful for therapeutic purposes. Instead of determining to which diagnostic category a particular child belongs, a more individualistic approach is advocated in which, for each particular speech-disordered child, all relevant underlying factors are assessed.

The aim of the present study is to evaluate and cross-validate the diagnostic procedure (Thoonen *et al.*, 1996), developed to assess motoric involvement in childhood speech disorders. In that study maximum performance tasks (MPT) (Kent, Kent and Rosenbek, 1987) were employed to quantify the speech motor capacities of children with dysarthria and developmental apraxia of speech (DAS). The results indicated that children with spastic dysarthria can be differentiated from both DAS and normal-speaking subjects on two MPT measures. Children with DAS, furthermore, could be differentiated from normal-speaking children mainly on the basis of their performance on trisyllabic repetition rate ('patakapataka . . .').

A considerable problem in the differentiation of underlying deficits is the variability of performances (Kent *et al.*, 1987) and the overlap in symptomatology between developmental conditions in general (Morgan Barry, 1995). For these reasons, in previous studies we used composite scores of comparable tasks, a procedure that greatly enhances the power of diagnostic differentiation (Thoonen *et al.*, 1996; Wit, Maassen, Gabreëls and Thoonen, 1993). Moreover, in these studies it is argued that for each of the speech mechanisms composite scores yield indices for degree of involvement. In the present study MPT, as used in Thoonen *et al.* (1996), are cross-validated against newly selected groups of children who can be considered less clear cases of dysarthria and DAS, due to milder involvement or more accompanying speech problems than in the previous study. This offers the opportunity to study the generalizability of the obtained results. Moreover, in the present study a group of children with non-specific speech disorders was added. For the latter group it remains to be determined to what extent MPT contribute to the assessment of underlying deficits.

Method

Subjects

Selection procedure

The selection procedure comprised three stages: referral, assessment/evaluation, group assignment, and two trajectories: an experimental protocol and a clinical protocol. The stages and trajectories are schematically displayed in figure 1. The following groups of subjects were selected. The *reference groups* consisted of the most clear cases of DAS and dysarthria. They were referred and recruited not primarily for a clinical diagnosis, but to serve as subjects in this and a previous study (Thoonen *et al.*, 1996). Children classified as cases with mixed disorders or milder involvement, based on the outcomes of experimental assessment, were assigned to the *validation groups*. Moreover, a group of children with non-specific articulation problems according to clinical assessment was added to the validation groups to study motoric involvement in this more heterogeneous group of children. These children were referred to a multidisciplinary diagnostic team at the University Hospital Nijmegen for diverse reasons. The validation groups were used to cross-validate the diagnostic procedure and criteria developed in a previous study (Thoonen *et al.*, 1996). For all children participating in this study, MPT were excluded as criteria for diagnostic classification in experimental as well as clinical assessment.

Reference groups

The reference groups were used in a previous study that addressed the clinical importance of MPT in differential diagnostic decision making in childhood motor speech disorders (Thoonen *et al.*, 1996). In the referral stage, children were selected at our request by the speech-language pathologists in schools. The most important inclusion criteria for DAS were high rates of speech sound errors, inconsistent error patterns, difficulties in producing articulatory complex sound sequences, and groping behaviour. For dysarthria the most important criteria were slow speech rate, imprecise consonants, and consistent error patterns. (For further details the reader is referred to Thoonen *et al.*, 1996.) The normal-speaking children were selected from regular primary schools by their classroom-teacher based on the following criteria: (a) no history of language or speech problems and (b) achieving average school performances.

During experimental assessment (left-hand panel of figure 1), all children were observed and tested to ascertain that each child complied with the following criteria: (a) normal range of intelligence, (b) adequate hearing capacities, (c) adequate receptive language abilities, (d) no organic abnormalities or otorhinolaryngological problems; and (e) a sufficient attention span. For speech assessment the following tasks were administered: imitation of words and nonsense words, imitation of sentences, imitation of pitch and temporal changes on phonation and spontaneous speech. These speech samples were analysed and results related to the diagnostic classification.

Experimental speech evaluation was performed by certified speech-language pathologists at the University Hospital Nijmegen, based on audiotaped recordings of spontaneous speech and sentence imitations. The speech criteria used for the classification were identical to those used in the referral stage. The consensus between judgements of the speech-language pathologists in diagnosis of the children was high (Thoonen *et al.*, 1996). In addition to the diagnostic classifications, severity

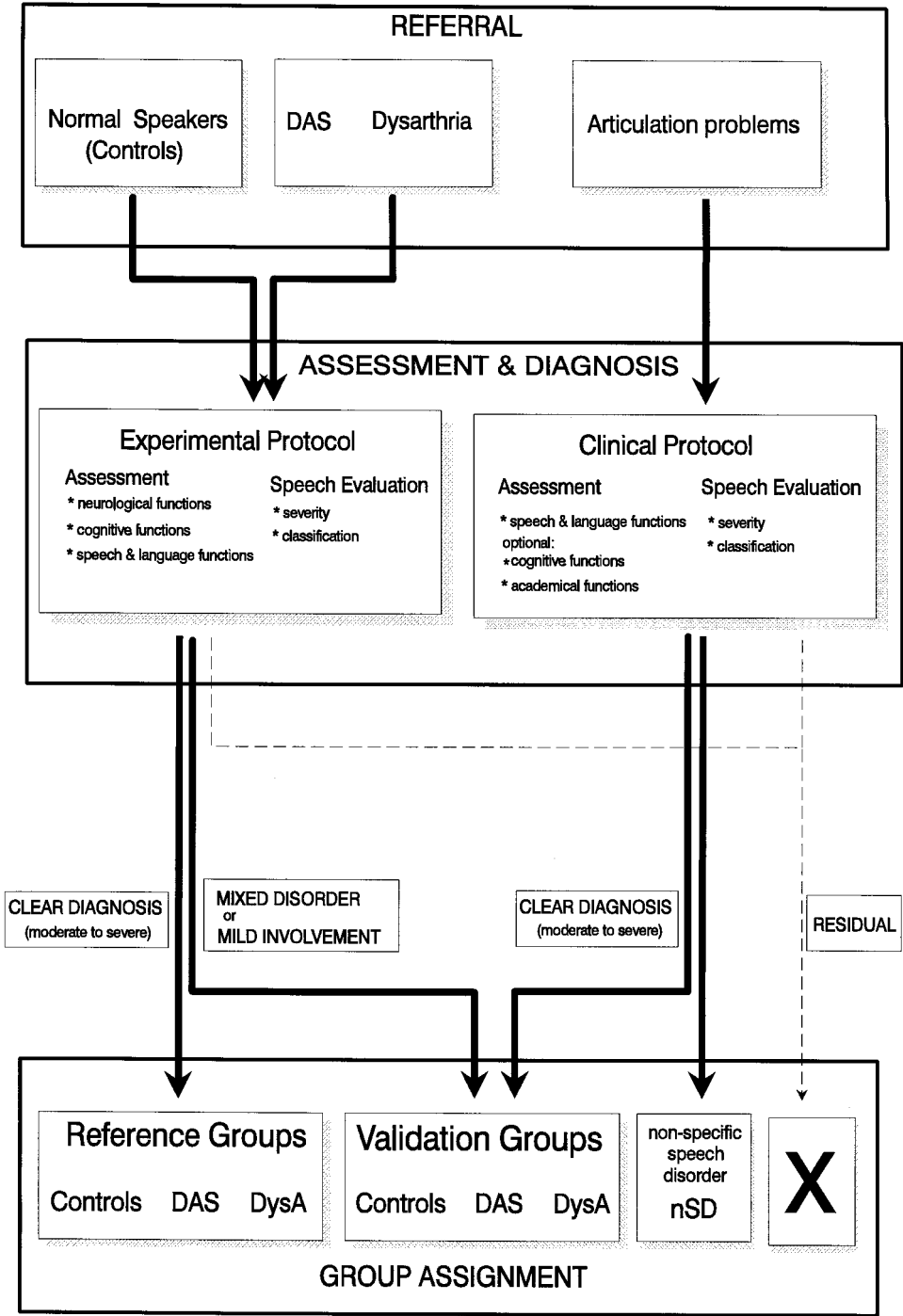


Figure 1. Protocolized experimental and clinical trajectories for group assignment.

ratings were also obtained, depicted on a four-point scale: normal speech (0), and mild (1), moderate (2), severe (3) involvement.

After experimental assessment and evaluation, a total number of 11 normal-speaking children, 11 children with a clear diagnosis of DAS and nine children with a clear diagnosis of spastic dysarthria were selected for the reference groups.

Validation groups

The validation groups consisted, firstly, of children who had gone through the selection procedure of the reference group (left-hand panel of figure 1), but who proved to have a mixed disorder or mild involvement according to speech evaluation of the experimental protocol. Six children entered the validation groups with primarily DAS and eight children with primarily dysarthria.

Added to the validation groups were children who were referred to the University Hospital Nijmegen for a differential diagnosis of their speech-language problems. During clinical assessment (right-hand panel of figure 1), several observations and tests were performed, including neurological and neuropsychological investigation (e.g. language and speech tasks). Besides clinical assessment, the speech-language problems of the subjects were evaluated by the speech-language pathologist involved. Classification and severity ratings were based on the same criteria that were applied for the selection of the reference groups. Only clinical cases with a clear diagnosis were admitted to the validation groups. The procedure of clinical assessment and evaluation resulted in a group of four children with DAS and one child with dysarthria.

After experimental and clinical assessment and evaluation, a total number of 10 children with DAS and nine children with spastic dysarthria were selected for the validation groups. A control group of 11 children was added in the same age range as the DAS and dysarthric group.

In comparison to the reference groups the validation groups are more heterogeneous with respect to (a) clarity in diagnosis, (b) severity and (c) age range. That is, children with mixed disorders or milder involvement, as well as adolescents (13–16 years) were accepted for the validation groups. The older children in the validation groups were all treated for their motoric (dysarthric) handicaps in revalidation centres.

Speech-disordered group

A group of 11 children with non-specific speech disorder was selected on the basis of the clinical assessment procedure. The term 'non-specific speech disorder' is used here to encompass children with articulation disorder, delayed phonological acquisition, consistent deviant disorder and inconsistent disorders (Dodd, 1995). It is a heterogeneous group with a speech disorder sufficiently severe to serve as the primary indication for special education and/or referral to a university diagnostic setting.

An overview of selected demographic characteristics for the reference groups, validation groups and the non-specific speech disordered group is shown in table 1.

Procedure

Maximum performance tasks

MPT were used to determine the upper limits of the speech-motor capacities of respiration, phonation, and articulation. A detailed description of the tasks and associated parameters is given in Thoonen *et al.* (1996). In short, one set of tasks (maximum sound prolongation) was enlisted to the protocols to evaluate the respirat-

Table 1. Subject characteristics of reference and validation groups: group size, age (in months) and gender

	Reference group				Validation group				Speech-disordered							
	Age		Gender		Age		Gender		Age		Gender					
	<i>n</i>	Mean	Range	M	F	<i>n</i>	Mean	Range	M	F	<i>n</i>	Mean	Range	M	F	Gender
Control	11	83.8	72–98	8	3	11	107.6	62–138	7	4						
DysA	9	97.7	76–123	8	1	9	137.7	64–197	6	3						
DAS	11	83.2	75–93	9	2	10	67.1	53–90	5	5						
nSD											11	93.1	52–131	10	1	

Note: Control = normal-speaking; DysA = dysarthria; DAS = developmental dyspraxia of speech; nSD = non-specific speech disorder.

ory and phonatory capacities. Parameters derived from these tasks were: maximum phonation duration (MPD) and maximum fricative duration (MFD). Another set of tasks (maximum repetition rate) was conducted to assess the coordinative and planning abilities to execute repetitive movements of the lips and tongue under stressed conditions of speech rate. The associated parameters were: maximum repetition rate of monosyllabic sequences (MRRmono) and trisyllabic sequences (MRRtri), sequencing skills (Sequence) and additional attempts required to produce a correct syllabic sequence (Attempts).

For the children going through the experimental assessment (i.e. reference groups and the majority of the children of the validation groups), MPT formed part of a larger, highly standardized assessment protocol, in which phonetic and phonological abilities were also tested. This experimental assessment took two or three sessions of 30 minutes. Children selected according to the clinical assessment (part of population of the validation groups) were presented with MPT as part of a neuropsychological and speech assessment.

On MPT, the children were stimulated to perform to the best of their capacities. To achieve this, each child received standardized instruction and a maximum of six trials to improve performance. The audiorecorded responses to the MPT were phonetically transcribed, qualified in terms of correctness (sound class, sequence) and subsequently selected for acoustic analysis. The maximum durations of sound prolongation (MSP) were measured directly from audiotape with a digital stopwatch. The maximum repetition rates (MRR) were interactively measured from the digitized signal with help of the Kay-Computerized Speech Lab model 4300B (for a detailed description of the procedure of measurement see Thoonen *et al.*, 1996).

Diagnostic flowchart and research parameters

In figure 2 a slightly adapted version of the flowchart that was the result of the previous study based on the reference groups (Thoonen *et al.*, 1996), is presented. In Appendix A the scores of all children are listed. In the current adapted flowchart, instead of combining parameters into composite scores, parameters are used separately. The main reason for this adaptation is that criteria based on single parameters are easier to apply in a clinical setting than the mathematically more complex, combined scores derived from *z*-scores. Below we will show that the two flowcharts are functionally equivalent.

One of the parameters in figure 2 is repetition rate of monosyllabic sequences (MRRmono). According to the flowchart, if a child produces a MRRmono below 3 syllables per second the child is diagnosed as dysarthric. In Appendix A the child obtains a 'dysarthria score' of 2. If the child produces a MRRmono of 3.5 syllables per second or faster, the child is diagnosed as *not* dysarthric (dysarthria score 0). In both these cases it is not necessary to administer the task MPD. However, if the child produces a MRRmono between 3.0 and 3.5 syllables per second (dysarthria score 1), the diagnosis also depends on the performance on MPD. If MPD is higher than 7.5 seconds, the dysarthria score remains 1 and the child is diagnosed as *not* (primarily) dysarthric. If, on the other hand, MPD is 7.5 seconds or below, the child obtains an additional dysarthria penalty score of 1, such that the total dysarthria score† becomes 2. A dysarthria score of 2 indicates that the child is primarily dysarthric.

† Because in the present study all tasks were administered to all children, instead of making MPD conditional to performance on MRRmono, or making MFD and Attempts conditional to Syllable Sequence and MRRtri, dysarthria and dyspraxia scores higher than 2 are reported in Appendix A.

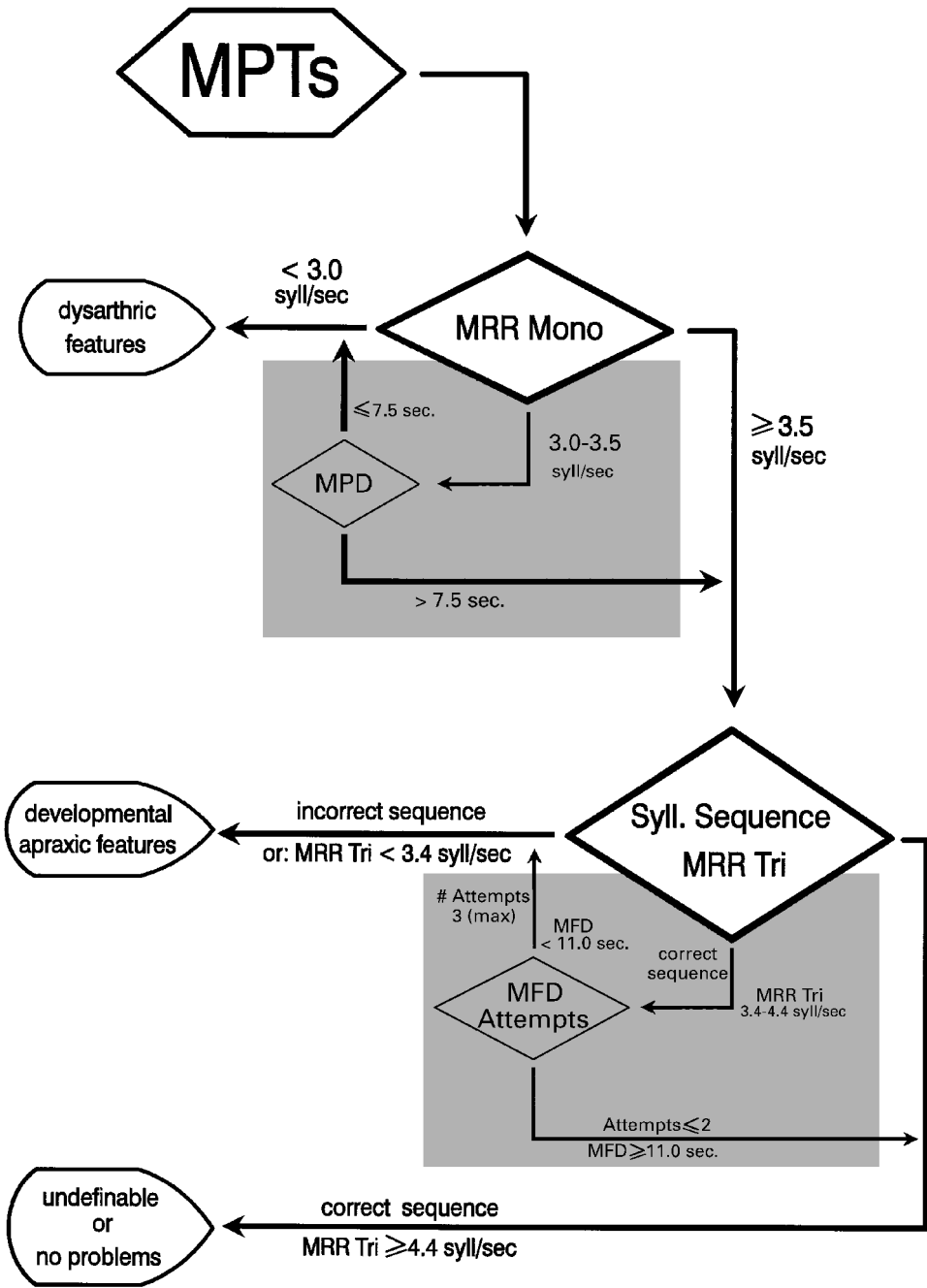


Figure 2. Adapted flow chart of the speech-motor model for differential diagnosis based on the weighted scores of five MPT parameters.

In the second stage of the flowchart of figure 2 the following parameters are used: (1) correctness of trisyllabic sequence (Syllable Sequence) and (2) repetition rate of a trisyllabic sequence (MRRtri). According to the criteria, the diagnosis

dyspraxia applies if a child is unable to produce a correct trisyllabic sequence or if a child produces a MRRtri below 3.4 syllables per second. In Appendix A the child obtains a dyspraxia score† of 2. If the child produces a correct trisyllabic sequence of 4.4 syllables per second or faster, the child is diagnosed as *not* dyspraxic (dyspraxia score 0). In these two cases no additional parameter(s) are required. However, if the child produces a MRRtri between 3.4 and 4.4 syllables per second (dyspraxia score 1), the diagnosis depends on MFD or the number of extra attempts (Attempts) the child required to achieve a correct trisyllabic sequence. If the child produces a MFD higher than 11 seconds, or needed less than 3 extra attempts, the dyspraxia score remains unaltered (1). The child is diagnosed as *not* (primarily) DAS. On the other hand, if MFD is below 11 seconds or Attempts is 3 (maximum), the child obtains an additional dyspraxia penalty score, such that the total dyspraxia score becomes 2. This dyspraxia score justifies the clinical diagnosis DAS.

The current, adapted flowchart of figure 2 is functionally equivalent to the flowchart presented in the previous study (Thoonen *et al.*, 1996). That is, all children in the reference groups obtained the same speech motor diagnosis in both flowcharts (i.e. the previous flowchart based on composites of parameters and the current flowchart with weighted parameters). Of the 9 children with dysarthria (according to experimental evaluation protocol) all were dysarthric according to both flowcharts. All 11 normal-speaking children were correctly diagnosed by both flowcharts. Only for the DAS reference group, one and the same child out of 11 children was missed according to both flowcharts.

Cross-validation

Evaluation of the predictive validity of MPT (i.e. the extent to which measures of MPT are able to classify groups of children using the adapted flowchart) consisted of computing sensitivity and specificity values‡ for the validation groups. The *sensitivity* of a diagnostic procedure expresses the accuracy of the procedure to detect individuals in a population who have the disorder. The *specificity* of a diagnostic procedure expresses the accuracy of the procedure to detect *only* disordered individuals (i.e. *not* diagnose healthy individuals as pathological).

Statistical analysis

In order to test whether the observed differences between validation groups were statistically significant, one-way analyses of variance (SPSS-PC; ANOVA, 1993) were conducted for each of the six MPT parameters under investigation. The main effect tested for significance was Validation Group (control, dysarthria, DAS). Measures of MRRmono and MRRtri were logarithmically transformed in order to normalize the positively skewed MRR distributions; the observed measures of skewness were $g > 0.40$. Pair-wise comparisons were made using the Newman-Keuls procedure ($p < 0.05$).

‡ Sensitivity and specificity are calculated on the basis of a 2×2 contingency table consisting of: (1) the number of children correctly diagnosed as having the disorder ('hits'); (2) the number of children incorrectly diagnosed as having the disorder ('false-positives'); (3) the number of children correctly diagnosed as *not* having the disorder ('correct rejections'); and (4) the number of children incorrectly diagnosed as *not* having the disorder ('false-negatives'). Sensitivity is calculated by the formula $(1)/(1)+(4)$ and specificity by the formula $(3)/(2)+(3)$.

Results

Dysarthria score: MRRmono and MPD

MRRmono

Table 2 shows that, for the validation groups, the dysarthric group produced slower monosyllabic sequences than the DAS group and control group. In a one-way analysis of variance on MRRmono the factor Subject Group (control, dysarthria, DAS) proved to be highly significant [$F(2,29)=47.18, p<0.001$]. A Newman–Keuls test revealed that the MRRmono for the dysarthric group was significantly lower than for the DAS and control groups ($p<0.05$). Furthermore, the DAS group produced significantly lower repetition rates than the controls. The performance pattern of the validation groups (DysA < DAS < Control) is different from that of the reference groups (DysA < DAS $\approx$ Control). As compared to the reference groups, children with DAS in the validation groups produced slower MRRmonos. We hypothesize that this difference is caused by co-occurring dysarthric involvement in the DAS validation group.

MPD

The mean maximum phonation durations of the validation groups are presented in table 2. A univariate analysis of variance on the MPD revealed significant differences between the three subject groups [$F(2,29)=11.59, p<0.01$]. A *post-hoc* test for group differences (Newman–Keuls) revealed that the MPD produced by the DAS and dysarthric groups were significantly shorter than that of the control group ($p<0.05$). The group means of the DAS and dysarthric groups were not statistically significant.

The performance pattern of the validation groups (DysA $\approx$ DAS < Control) is different from that of the reference groups (DysA < DAS $\approx$ Control). In particular, the MPD of the dysarthric validation group were significantly higher than that of the dysarthric reference group [$t=3.32, \text{d.f.}=16, p=0.004$]. The observed differences between the dysarthric validation and reference groups can be attributed to normal levels of MPD for four children in the higher age range with mixed forms of dysarthria (Appendix B: cases DysA 12 to DysA 15). Moreover, the dyspraxic validation group performed significantly poorer on MPD than the dyspraxic reference group [$t=-2.63, \text{d.f.}=19, p=0.02$]. Inspection of the individual data for the DAS groups indicated that the MPD clearly improves between the ages 4–5 years and 6–8 years [Spearman correlation coefficient: $r(\text{MPD}, \text{age})=0.62, n=21, p<0.01$]. No trends reflecting age-effects were found in MPD of the dysarthric groups.

The scatterplots for the reference and validation groups (see figure 3) illustrate the coherence between MRRmono and MPD for three subject groups. The uninterrupted vertical line in both plots serves as a distinctive marker for dysarthria scores below 2 or equal/above 2. This boundary underscores the importance of MRRmono for differential diagnosis between the validation groups. Although the differences between the validation and reference groups are most evident on MPD, MRRs in the critical bandwidth (3.0–3.5) always co-occur with short MPD. This observation emphasizes the importance of MPD, in addition to MRRmono, for the differentiation of dysarthric symptomatology from other speech problems. Inspection of the individual data reveals that children from the dyspraxic validation group achieve

Table 2. Maximum repetition rate of monosyllabic sequences (MRRmono in syllables/second) and maximum phonation duration (MPD in seconds) for the validation groups

Parameter	Validation groups									
	Control			Dysarthria			DAS			nSD
	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range	
MRR mono	5.27	0.57	4.37–6.14	2.74	0.83	1.83–4.27	3.80	0.23	3.46–4.22	4.30
MPD	14.98	2.72	10.73–20.70	10.31	5.49	4.01–18.09	7.10	2.78	3.86–13.10	10.52
										2.71–16.36

Note: Control= normal-speaking [n = 11]; Dysarthria [n = 9]; DAS= developmental dyspraxia of speech [n = 10]; nSD= nonspecific speech disorder [n = 11].

Dysarthria Score

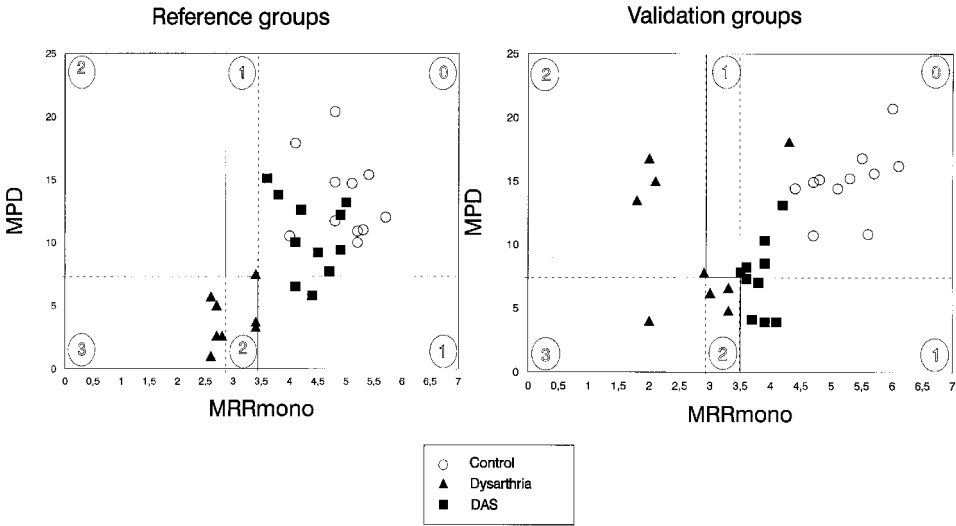


Figure 3. Scatterplots of *MRRmono* and *MPD* for the reference groups and the validation groups. Note. The encircled values represent the summarized penalty scores on *MRRmono* and *MPD*. *MPD* below 7.0 seconds and *MRRmono* between 3.0–3.5 syllables/second receive a penalty score of 1, whereas *MRRmono* below 3.0 receive a penalty score of 2. (a) Dysarthria score=0 (no penalty scores on *MRRmono* and *MPD*); (b) dysarthria score=1 (one penalty score on *MRRmono* or *MPD*); (c) dysarthria score=2 (one penalty score on *MRRmono* and *MPD* or two penalty scores on *MRRmono*); (d) dysarthria score=3 (two penalty scores on *MRRmono* and one penalty score on *MPD*).

lower scores on both *MRRmono* and *MPD* in comparison to children from the dyspraxic reference group. Normal-speaking children from the validation group perform slightly better on both parameters.

Dyspraxia score: Sequence, Attempts, MRRtri and MFD

Sequence and attempts

Prior to the calculation of trisyllabic repetition rates (*MRRtri*), the performances were evaluated upon correctness of the sequence produced by the child (Sequence) and upon ease of production (Attempts). Table 3 shows the number of children in the validation groups capable of producing a correct trisyllabic sequence, as well as the number of children who require no, one, two or three (= maximum) extra attempts.

All normal-speaking children were able to produce fast, error-free trisyllabic sequences (see also Appendix B). Eight out of 11 normal-speaking children, and six out of nine dysarthric children, were able to produce a trisyllabic sequence immediately upon standard practice (i.e. required no extra attempts). Three normal-speaking and three dysarthric children required extra attempts. One dysarthric child turned out to be incapable of producing a correct trisyllabic sequence. In the dyspraxic validation group, only 2/10 children were able to produce a correct trisyllabic sequence, and both children needed extra attempts.

Table 3. *Distribution of subjects on sequence (ability of producing a trisyllabic sequence) and Attempts (number of extra attempts required upon general practice for the production of a trisyllabic sequence) for the validation groups*

	Sequence		Attempts		
	Correct	Incorrect	0	1–2	3
Control	11	0	8	2	1
Dysarthria	8	1	6	1	2
DAS	2	8	0	1	9
nSD	9	2	2	7	2

Note: Control= normal-speaking [$n=11$]; DysA= dysarthria [$n=9$]; DAS= developmental dyspraxia of speech [$n=10$]; nSD= non-specific speech disorder [$n=11$].

The performance patterns of the dysarthric validation and reference groups do not differ. For the dyspraxic groups, however, more children in the reference group (7/11) were capable of producing a trisyllabic sequence, as compared to 2/10 children in the validation group. This difference might be caused by the fact that more younger children participated in the dyspraxic validation group than in the dyspraxic reference group. The data of the dyspraxic reference and validation groups indicate that children under the age of 6 years experience severe difficulties in producing a correct trisyllabic sequence of /pataka/.

MRRtri

Performances on Maximum Trisyllabic Repetition Rate (MRRtri) and Maximum Fricative Duration (MFD) are presented in table 4. Maximum repetition rates were only calculated for error-free trisyllabic sequences. As a result of this restriction, only for two dyspraxic children could trisyllabic rates be calculated. A one-way analysis of variance on MRRtri yielded highly significant differences between the three subject groups (i.e. control, dysarthria and DAS) [$F(2,20)=18.67, p<0.001$]. Newman–Keuls *post-hoc* comparisons ($p<0.05$) indicated that the control group produced higher trisyllabic rates than the DAS and dysarthric groups; the latter groups did not differ. This performance pattern (DAS $\approx$ DysA < Control) of the validation groups is identical to that of the reference groups. Moreover, a comparison of the validation and reference groups revealed no significant differences for the dyspraxic groups [$t=-2.24, \text{d.f.}=7, p=0.06$] and for the dysarthric groups [$t=1.15, \text{d.f.}=14, p=0.271$].

MFD

A one-way analysis on the MFD for the validation groups indicated significant differences between subject groups [$F(2,30)=8.98, p<0.01$]. A Newman–Keuls test ($p<0.05$) demonstrated significant differences between the control group and the other two groups, which did not differ. This results in the following performance pattern: DAS $\approx$ DysA < Control. In the reference groups (see Appendix A), the children with dysarthria produced significantly shorter MFD than the children with DAS. These results can be translated into the performance pattern: DysA < DAS < Control. In correspondence to MPD, the observed differences between validation and reference groups can be attributed to normal-like behaviour of some older dysarthric children and to dysarthric-like behaviour of some younger children with DAS.

Table 4. *Maximum repetition rate of trisyllabic sequence (MRRtri in syllables/second) and maximum fricative duration (MFD in seconds) for the validation groups*

Parameter	Validation groups											
	Control			Dysarthria			DAS			nSD		
	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range
MRR tri	5.33	0.96	3.77–7.06	2.93 ^a	0.99	1.33–4.48	2.44 ^b	0.02	2.42–2.45	3.84 ^c	1.20	2.56–5.71
MFD	15.01	3.31	9.99–20.93	10.13	5.72	3.48–17.77	6.76	3.39	2.01–17.10	10.72	3.84	4.11–17.82

Note: Control= normal-speaking [$n = 11$]; Dysarthria [$n = 9$]; DAS= developmental dyspraxia of speech [$n = 10$]; nSD= non-specific speech disorder [$n = 11$]

Children unable to produce a correct trisyllabic sequence after a maximum number of additional trials:

^aOne of nine cases excluded (11.1%).

^bEight of 10 cases excluded (80.0%).

^cTwo of 11 cases excluded (18.2%).

Evaluation of diagnostic flowchart

The outcome of the diagnostic flowchart of figure 2 for the validation groups (total number of 41 children) was calculated against the clinical judgements of speech-language pathologists. Sensitivity and specificity of the diagnostic procedure are shown in table 5. The first 2×2 contingency table demonstrates that eight of the nine children (89%) clinically judged as dysarthric obtained a dysarthric score equal to or higher than 2. Thus, application of the two MPT (i.e. MRRmono and MPD) yields a sensitivity of 89%, which suggests that MRRmono and MPD are accurate measures to detect children with moderate to severe dysarthria. Furthermore the diagnostic procedure proved to be highly specific: none of the children with other clinical labels was classified as dysarthric.

A similar result was obtained for the children with DAS. All children in the validation group with a primary clinical diagnosis of DAS received a dyspraxia score of 2 or higher. This means that the sensitivity of the flowchart (applying Sequence, MRRtri, MFD and Attempts) to detect DAS is 100%. However, one normal-speaking child received a dyspraxia score of 2 and thus was classified wrongly; this means that the specificity of the flowchart is 91% (10/11). Information from the classroom teacher revealed that, for this incorrectly classified normal-speaking child, some minor speech problems were perceived in comparison to peers. Referral for speech therapy was not yet considered, however.

Table 5. *Contingency tables of validation measures (sensitivity and specificity) of classifications (dysarthria and dyspraxia) based on MPT_s (adapted flowchart) as compared to judgements of speech-language pathologists*

Classification of dysarthria
MRRmono and MPD

Diagnosis	Speech-language pathologist		Total
	Dysarthria	Not dysarthria	
MPT			
Dysarthria	8	0	8
Not dysarthria	1	21	22
Total	9	21	30

Sensitivity: 89% (8/9); specificity: 100% (21/21).

Classification of dyspraxia
Sequence, MRRtri, MFD and Attempts

Diagnosis	Speech-language pathologist		Total
	DAS	Not DAS	
MPT			
DAS	10	1	11
Not DAS	0	10	10
Total	10	11	21

Sensitivity: 100% (10/10); specificity: 91% (10/11).

Non-specific speech-disordered group

Performance patterns on MPT

The performances of the non-specific speech-disordered group on the MPT parameters were compared to the performances of the validation groups. On MRRmono the mean repetition rates of the speech-disordered group lie in between that of the control and DAS group (see table 2). Based on the results of a *post-hoc* analysis (Newman–Keuls, $p < 0.05$) the following pattern emerged: $\text{DysA} < \text{DAS} \approx \text{nSD} < \text{Control}$. For MPD the performances of the speech-disordered group were similar to that of the DAS and dysarthric groups ($\text{DysA} \approx \text{DAS} \approx \text{nSD} < \text{Control}$). The observed differences between the speech-disordered and DAS group (10.52 versus 7.10) were not statistically significant [$t = 2.05$, d.f. = 19, $p = 0.054$].

Most of the children with non-specific speech disorders were capable of producing a rapid, error-free trisyllabic sequence (Sequence). Regarding this outcome, the speech-disordered group strongly resembles the control and dysarthric groups. However, in contrast to the latter groups, the speech-disordered group required more extra attempts (one or two) upon standard practice (see table 3). This performance profile is quite different from that of all the other subject groups in this study. On MRRtri the children with non-specific speech disorders performed within a wider range of repetition rates (see table 4) in comparison to the other validation groups. The mean repetition rates of the speech-disordered group are significantly slower than that of the control group [$t = -3.09$, d.f. = 18, $p < 0.001$]. The following performance pattern emerges: $\text{DAS} \approx \text{DysA} \approx \text{nSD} < \text{Control}$. The mean performances of the non-specific speech-disordered group on MFD are similar to the dysarthric group. Furthermore, for both groups the diversity in MFDs is quite extensive (ranges from 4 to 18 syllables/second). The speech-disordered group produced significantly shorter MFD than the control group [$t = -2.81$, d.f. = 20, $p < 0.01$] but significantly longer MFD than the DAS group [$t = 2.21$, d.f. = 19, $p < 0.04$]. Thus, the following performance pattern emerges: $\text{DAS} < \text{DysA} \approx \text{nSD} < \text{Control}$.

Differential diagnosis based on flowchart

The outcome of the diagnostic procedure of MPT (see flowchart in figure 2) for the non-specific speech-disordered group is shown in table 6. This table demonstrates that one of the 11 children (9%), clinically judged as non-specific speech-disorder, obtained a dysarthric score equal to 2. For this child (Appendix B: nSD no. 1) the diagnosis of moderate to severe dysarthria is therefore justified. This child also obtained a dyspraxia score of 2. An additional six of the remaining 10 children (60%) were classified as primarily dyspraxic, based on dyspraxia scores of 2 or higher. The speech–language problems of four children (36%) could not be characterized as originating from speech-motor dysfunctions. These four children obtained a dysarthria and dyspraxia score of 0 or 1.

Discussion

In a previous study we reported on a diagnostic procedure, based on maximum performance tasks (MPT), for the assessment of motoric involvement in children with speech problems. In the present study we aimed at evaluating the clinical validity of this procedure, as well as the differential diagnostic power, using newly selected cases with pure and mixed speech–language disorders of moderate severity.

Table 6. *Distribution of dysarthria scores and dyspraxia scores for the validation groups*

	Dysarthria score		Dyspraxia score	
	< 2	≥ 2	< 2	≥ 2
nSD	10 (91%)	1 (9%)	4 (36%)	7 (54%)
Control	11 (100%)	0 (0%)	11 (100%)	0 (0%)
Dysarthria	1 (11%)	8 (89%)	2 (22%)	7 (78%)
DAS	10 (100%)	0 (0%)	0 (0%)	10 (100%)

Note: nSD= non-specific speech disorder [$n=11$]; Control= normal-speaking [$n=11$]; Dysarthria [$n=9$]; DAS= developmental dyspraxia of speech [$n=10$]

The main aim of the present study was to cross-validate a diagnostic procedure based on MPT for the assessment of motoric involvement in speech problems of school-aged children. Validation of the diagnostic procedure (decision rules and criteria) was performed on subject samples (validation groups) that were selected independently of the subject samples (reference groups) used to create the decision rules and criteria in the first place. The results indicate that the validity, expressed in sensitivity and specificity, of the diagnostic procedure was high when compared to clinical judgements. The sensitivity measures were 89% for dysarthria and 100% for dyspraxia. The overall diagnostic accuracy (sensitivity) of the procedure was 95%. Moreover, the overall specificity of the procedure was very high (97%): only one of the 30 (3%) participating children was misdiagnosed. In clinical practice, sensitivity (true-positives) and specificity (true-negatives) values of at least 90% are required (Shriberg, Aram and Kwiatkowski, 1997). Thus, the results clearly support the validity of the diagnostic procedure in the classification of children with motor-speech disorders. More specifically, this implies that subsets of MPT measures yield indices of speech-motor behaviour that have clear differential diagnostic value. The high sensitivity is particularly notable because this study utilized six (30%) speech-disordered children with only mild involvement or indefinite diagnosis. In other words, the admittance of less clear cases had no negative impact on the sensitivity of the diagnostic procedure.

A review of the performance patterns of the validation and reference groups revealed that a slow monosyllabic repetition rate (MRRmono) is a powerful sign of dysarthric involvement. Although the dysarthric cases of the validation and reference groups show rather diverse performance patterns on MRR (monosyllabic repetition rate versus trisyllabic repetition rate), and thus represent rather heterogeneous subject groups, MRRmono remains the prime diagnostic marker for spastic dysarthria. This finding corroborates the conclusion of the previous study (Thoonen *et al.*, 1996). Also in line with the latter study is the role of maximum phonation duration (MPD) as a contributing parameter in the diagnosis of dysarthria. The present study demonstrates that MPD is also an essential parameter for a proper diagnosis of less clear cases of dysarthria. Two children achieved intermediate MRRmono values (3.0–3.5), and could therefore not be diagnosed on merely one parameter. In general, the results of this study suggest that MPD is less suitable, in terms of sensitivity, for young children under 4 years of age.

In correspondence with the findings based on the reference groups, difficulty in sequencing speech movements is the central sign of dyspraxic involvement. The MPT measures, ease of production (Attempts) and trisyllabic repetition rates (MRRtri) proved to be very sensitive measures to assess speech planning capacities

in school-aged children. The status of maximum fricative duration (MFD) for the diagnosis of dyspraxia remains undetermined, because the unambiguous performances on MRRtri for the dyspraxic validation group were highly informative towards making a diagnosis. As yet, MFD will be retained as a supporting measure for a proper diagnosis of less clear cases of dyspraxia.

Regarding the specificity value of sequencing abilities, the relatively poor performance of one normal-speaking child (Appendix B: C15) was validated by the recognition of speech-language problems nearly 1 year after the assessment of MPT. The results of this study also indicated that 4–5-year-old children with dyspraxia will hardly be able to produce a correct sequence, whereas normal-speaking age mates require more practice but ultimately grasp the task in producing relatively slow repetition rates (Logan, Yaruss and Conture, 1994).

The second aim of the present study concerns the contribution of the diagnostic procedure of MPT in the differentiation of non-specific speech disorders from dysarthria and dyspraxia. The results of this part of the study indicate that the performance pattern of the non-specific speech-disordered group is different from that of remaining subject groups. Firstly, children from the non-specific speech-disordered group required more practice to produce trisyllabic sequences than children from the dysarthric and control groups. Secondly, in contrast to the dyspraxic group a large proportion of the non-specific speech-disordered group is capable of producing a correct trisyllabic sequence after prolonged practice. On MRRmono and MPD the performances of the non-specific speech-disordered group are intermediate between that of the control and DAS validation groups, but closest to the performances of the DAS group. On MRRtri and MFD the performances of the non-specific speech-disordered group are closest to that of the dysarthric group.

The results show that at the level of speech symptoms, expressed in MPT scores, there is an overlap in performances across the three subject groups. To illustrate this, the study revealed: (a) a co-occurring dysarthric involvement in the dyspraxic validation group as assessed by MRRmono; (b) a normal-like behaviour of some older dysarthric cases on MPD and MFD; and (c) a dysarthric-like behaviour in the youngest children with DAS on MFD. Moreover, the MPT measures indicated that a dyspraxic or dysarthric involvement is often present in children with a non-specific speech disorder. In spite of the overlap in speech symptomatology, MPT yielded accurate indices of the primary speech deficits for each particular child. From a clinical perspective the assessment of the relative involvement of dyspraxia versus dysarthria in speech production requires the administration of both MRRmono and MRRtri. The combination of results from MRRmono and MRRtri not only yields a diagnostic classification, but also determines the degree of involvement of secondary speech disturbances. With these results the study has substantiated the generalizability of the diagnostic procedure, and thereby pointed out its use in clinical (diagnostic) practice.

This study has clearly shown that measures of MPT contribute to the differential diagnosis and assessment of childhood speech disorders. Moreover, only recently, several MPT have been automatically analysed by using computerized procedures (CSL-Motor Speech Profile, 1997; Bakker, Maassen and Thoonen, 1994; Thoonen, Bakker and Maassen, 1990). These customized procedures, using sets of protocols which quickly extract the speech parameters relevant for DAS and dysarthria, largely enhance the clinical relevance. This means that MPT are not only valid measures in speech-motor assessment, but are also relatively easy to manage in assessment as well as analysis.

Acknowledgements

The Netherlands Organization for Scientific Research (NWO) is gratefully acknowledged for funding of this project. This research was conducted while the first author was supported by a grant from the Psychology and Psychonomics Research Foundation (PSYCHON) of this organization (560-256-043) awarded to the second author.

References

- BAKKER, K., MAASSEN, B. and THOONEN, G., 1994, *Automated assessment of diadochokinetic performance: its accuracy, validity, and reliability*. (Mini-seminar presented at the American Speech–Language–Hearing Association Convention, New Orleans, LA.)
- BRADFORD, A. and DODD, B., 1996, Do all speech-disordered children have motor deficits? *Clinical Linguistics and Phonetics*, **10**, 77–102.
- DODD, B., 1995, *Differential Diagnosis and Treatment of Children with Speech Disorder* (London: Whurr).
- ENDERBY, P. and PHILIPP, R., 1986, Speech and language handicap: towards knowing the size of the problem. *British Journal of Disorders of Communication*, **21**, 151–165.
- KAY COMPUTERIZED SPEECH LAB (CSL), 1997, *Motor Speech Profile* (Pine Brook, NJ: Kay Elemetrics Corp.).
- KENT, R. D., KENT, J. F. and ROSENBEK, J. C., 1987, Maximum performance tests of speech production. *Journal of Speech and Hearing Research*, **52**, 367–387.
- LOGAN, K. J., YARUSS, J. S. and CONTURE, E. G., 1994, *Diadochokinetic assessment with young children: Options for measurement and interpretation*. (Mini-seminar presented at the American Speech–Language–Hearing Association Convention, New Orleans, LA.)
- MORGAN BARRY, R., 1995, The relationship between dysarthria and verbal dyspraxia in children: a comparative study using profiling and instrumental analysis. *Clinical Linguistics and Phonetics*, **9**, 277–309.
- SCHWARTZ, R. G., 1992, Clinical applications of recent advances in phonological theory. *Language, Speech, and Hearing Services in Schools*, **23**, 269–276.
- SHRIBERG, L. D., 1994, Five subtypes of developmental phonological disorders. *Clinics in Communication Disorders*, **4**, 38–53.
- SHRIBERG, L. D. and KWIATKOWSKI, J., 1982, Phonological disorders. I: A diagnostic classification system. *Journal of Speech and Hearing Disorders*, **47**, 226–241.
- SHRIBERG, L. D., ARAM, D. M. and KWIATKOWSKI, J., 1997, Developmental apraxia of speech: I. Descriptive and theoretical perspectives. *Journal of Speech and Hearing Research*, **40**, 273–285.
- SHRIBERG, L. D., KWIATKOWSKI, J., BEST, S., HENGST, J. and TERSELIC-WEBER, B., 1986, Characteristics of children with phonological disorders of unknown origin. *Journal of Speech and Hearing Disorders*, **51**, 140–161.
- THOONEN, G., BAKKER, K. and MAASSEN, B., 1990, Computerized determination of diadochokinetic rate and durational variability among dyspraxic and normal-speaking children. (Submitted).
- THOONEN, G., MAASSEN, B., WIT, J., GABREËLS, F. and SCHREUDER, R., 1996, The integrated use of maximum performance tasks in differential diagnostic evaluations among children with motor speech disorders. *Clinical Linguistics and Phonetics*, **10**, 311–336.
- WIT, J., MAASSEN, B., GABREËLS, F. and THOONEN, G., 1993, Maximum performance tests in children with developmental spastic dysarthria. *Journal of Speech and Hearing Research*, **36**, 452–459.

Appendix A

Individual data for reference groups over five MPT parameters with the combined weighted scores of parameters MRR mono and MPD and of parameters Sequence, MRR tri, Attempts and MFD

Subject group	Case	Age (months)	MRR- mono	MPD	Dysarthria score ^a	MRR- tri	Attempts (3 = maximum)	MFD	Dyspraxia score ^b
Control (<i>n</i> = 11)	C1	072	5.2	10.9	0	5.6	3	20.2	1
	C2	072	5.2	10.0	0	4.3	0	12.7	1
	C3	077	4.8	14.8	0	4.7	0	11.4	0
	C4	079	4.8	11.7	0	5.6	1	13.8	0
	C5	083	4.0	10.5	0	4.1	0	12.8	1
	C6	084	4.1	17.9	0	4.8	1	13.4	0
	C7	085	5.3	11.0	0	5.0	2	10.2	1
	C8	087	5.1	14.7	0	4.5	0	19.0	0
	C9	089	4.8	20.4	0	4.2	1	16.9	1
	C10	095	5.4	15.4	0	5.4	0	17.2	0
	C11	098	5.7	12.0	0	6.4	0	16.0	0
Dysarthria (<i>n</i> = 9)	DysA1	076	3.4	3.3	2	1.5	0	3.2	3
	DysA2	081	2.6	5.7	3	Unable	3	2.3	3
	DysA3	081	3.4	3.3	2	3.3	0	3.4	3
	DysA4	092	2.6	1.0	3	1.9	0	0.9	3
	DysA5	093	3.4	7.5	2	3.1	0	5.1	3
	DysA6	094	3.4	3.7	2	3.1	0	3.0	3
	DysA7	118	2.8	2.6	3	2.6	0	3.7	3
	DysA8	121	2.7	5.0	3	2.3	2	3.9	3
	DysA9	123	2.7	2.6	3	1.6	0	1.9	3
	DAS1	075	4.5	9.2	0	3.0	2	14.0	2
	DAS2	077	4.1	6.5	1	Unable	3	8.2	3
DAS (<i>n</i> = 11)	DAS3	078	3.6	15.1	0	4.0	3	10.8	2
	DAS4	079	4.2	12.6	0	3.7	1	9.1	2

Subject group	Case	Age (months)	MRR-mono	MPD	Dysarthria score ^a	MRR-tri	Attempts (3 = maximum)	MFD	Dyspraxia score ^b
	DAS5	081	4.1	10.0	0	Unable	3	7.9	3
	DAS6	082	4.4	5.8	1	Unable	3	7.5	3
	DAS7	084	4.7	7.7	0	3.1	1	11.7	2
	DAS8	085	3.8	13.8	0	2.8	1	11.6	2
	DAS9	089	5.0	13.2	0	4.7	1	11.2	0
	DAS10	092	4.9	9.4	0	4.3	3	6.2	2
	DAS11	093	4.9	12.2	0	Unable	3	6.3	3

^aDysarthria score:

Parameter	0	Weighted score	1	2
MRRmono	≥ 3.5	3.0–3.5		< 3.0
MPD	> 7.5	< 7.5		

^bDyspraxia score:

Parameter	0	Weighted score	1	2
Sequence	able			unable
MMR tri	≥ 4.4	3.4–4.4		< 3.4
Attempts	≤ 2	> 2		
MFD	≥ 11.0	< 11.0		

Copyright of Clinical Linguistics & Phonetics is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.